further course of which is still hidden in darkness. For now the indeterminists are justified in raising the question whether the concept of a causal influence of the measuring instrument on the measuring process has any intelligible meaning, since every attempt to examine such a causal effect directly requires some kind of measurement, and since with every new measurement a new causal influence and therefore a new uncertainty would be introduced into the problem.

And yet, this objection still does not dispose of the matter. For as every experimental physicist knows, besides the direct methods of investigation there exist also indirect ones, and the latter accomplish good results in many a case where the former have failed. However, I must take exception to the view (a very popular one these days, and certainly a very plausible one on the face of it) that a problem in physics merits examination only if it is established in advance that a definite answer to it can be obtained. If physicists had always been guided by this principle, the famous experiment of Michelson and Morley undertaken to measure the so-called absolute velocity of the earth, would never have taken place, and the theory of relativity might still be nonexistent. Accordingly, if the study of a question now regarded fairly universally as meaningless, such as that of the absolute velocity of the earth, has turned out to produce such extraordinary benefits to science, how much more worth while must it be to follow up a problem, the deeper meaning of which is still under debate and which is capable more than any other of enriching research.

But how are we to come to a decision? Obviously, there is no other alternative than to consider the two opposing views, to side with the one that appeals to us more, and then to investigate whether it leads to valuable or to worthless conclusions. At any rate, one must welcome the fact that physicists who are closely interested in this subject are split into two factions; one leans toward the theory of determinism, the other toward indeterminism. So far as I can see, the latter represent the majority nowadays, although it is difficult to establish the facts which may easily change in the course of time. Between these two viewpoints there is also room for a third position which in a certain sense is a mediating one, in that it assigns to certain concepts, such as the force of electrical attraction, a direct significance and a strict uniformity with respect to the sense world; whereas to other concepts, such as material waves, it assigns merely a statistical significance. However, because it lacks systematic unity this view does not appear to be very satisfactory. For this reason, I propose to disregard it now and to confine myself to a somewhat closer analysis of the two fully consistent viewpoints.

The indeterminist is satisfied in his quest for knowledge by the discovery that the wave function of quantum mechanics is merely a probability value. He has no further problem in connection with it. On the other hand, he sees